

A Theory of Components and Specifications

Karl Palmskog

This note presents a formal language of terms and predicates. We first define the language syntax and then provide its semantics and (incomplete) proof system.

Syntax

c	$::=$	component term
	$\mathbf{c}$	constant
	$c \times c'$	composition
S	$::=$	specification term
	$\mathbf{S}$	constant
	$S \sqcap S'$	conjunction
	(S, S')	assume-guarantee
	$S S'$	parallel
P	$::=$	predicate
	$c : S$	implements
	$S \sqsubseteq S'$	refines
	$P \rightarrow P'$	implies
	$P \wedge P'$	conjunction
	$P \vee P'$	disjunction
	$\neg P$	negation

Semantics

The semantics of component terms are given by a mapping of syntax to “behaviors” B from a set $\mathbb{B}$. The semantics of specification terms are given by a mapping of syntax to *sets* of behaviors, i.e., to subsets of $\mathbb{B}$. The meaning of predicates is given by mapping of syntax to meta-language propositions.

The meaning of constants is given by models $\mathcal{M}$ that assign them to behaviors or sets of behaviors. We assume there is a binary operator $\star$ on behaviors that returns a behavior, representing semantic composition.

Semantics of component terms:

$$\begin{aligned}\llbracket \mathbf{c} \rrbracket_{\mathcal{M}} &= \mathcal{M}(\mathbf{c}) \\ \llbracket c_1 \times c_2 \rrbracket_{\mathcal{M}} &= \llbracket c_1 \rrbracket_{\mathcal{M}} \star \llbracket c_2 \rrbracket_{\mathcal{M}}\end{aligned}$$

Semantics of specification terms:

$$\begin{aligned}\llbracket \mathbf{S} \rrbracket_{\mathcal{M}} &= \mathcal{M}(\mathbf{S}) \\ \llbracket S_1 \sqcap S_2 \rrbracket_{\mathcal{M}} &= \llbracket S_1 \rrbracket_{\mathcal{M}} \cap \llbracket S_2 \rrbracket_{\mathcal{M}} \\ \llbracket (S_1, S_2) \rrbracket_{\mathcal{M}} &= \{B \in \mathbb{B} \mid \forall B' \in \llbracket S_1 \rrbracket_{\mathcal{M}}. B \star B' \in \llbracket S_2 \rrbracket_{\mathcal{M}}\} \\ \llbracket S_1 \parallel S_2 \rrbracket_{\mathcal{M}} &= \{B \in \mathbb{B} \mid \exists B_1 B_2. B = B_1 \star B_2, B_1 \in \llbracket S_1 \rrbracket_{\mathcal{M}}, B_2 \in \llbracket S_2 \rrbracket_{\mathcal{M}}\}\end{aligned}$$

Semantics of predicates:

$$\begin{aligned}\llbracket c : S \rrbracket_{\mathcal{M}} &\Leftrightarrow \llbracket c \rrbracket_{\mathcal{M}} \in \llbracket S \rrbracket_{\mathcal{M}} \\ \llbracket S_1 \sqsubseteq S_2 \rrbracket_{\mathcal{M}} &\Leftrightarrow \llbracket S_1 \rrbracket_{\mathcal{M}} \subseteq \llbracket S_2 \rrbracket_{\mathcal{M}} \\ \llbracket P_1 \rightarrow P_2 \rrbracket_{\mathcal{M}} &\Leftrightarrow \llbracket P_1 \rrbracket_{\mathcal{M}} \text{ implies } \llbracket P_2 \rrbracket_{\mathcal{M}} \\ \llbracket P_1 \wedge P_2 \rrbracket_{\mathcal{M}} &\Leftrightarrow \llbracket P_1 \rrbracket_{\mathcal{M}} \text{ and } \llbracket P_2 \rrbracket_{\mathcal{M}} \\ \llbracket P_1 \vee P_2 \rrbracket_{\mathcal{M}} &\Leftrightarrow \llbracket P_1 \rrbracket_{\mathcal{M}} \text{ or } \llbracket P_2 \rrbracket_{\mathcal{M}} \\ \llbracket \neg P \rrbracket_{\mathcal{M}} &\Leftrightarrow \text{not } \llbracket P \rrbracket_{\mathcal{M}}\end{aligned}$$

Proof System

We give a proof system for predicates as a relation $\Gamma \vdash P$, where Γ is a set of predicates—intuitively the premises used for proving P .

$$\begin{array}{c} \overline{\Gamma, P \vdash P} \quad \text{AX} \\[10pt] \frac{\Gamma \vdash c : S_1 \quad \Gamma \vdash S_1 \sqsubseteq S_2}{\Gamma \vdash c : S_2} \quad \text{REF_EL} \\[10pt] \frac{\Gamma \vdash c : S_1 \quad \Gamma \vdash c : S_2}{\Gamma \vdash c : S_1 \sqcap S_2} \quad \text{CONJ_IN} \end{array}$$

$$\begin{array}{c}
\frac{\Gamma \vdash c : S_1 \sqcap S_2}{\Gamma \vdash c : S_1} \quad \text{CONJ_EL1} \\
\frac{\Gamma \vdash c : S_1 \sqcap S_2}{\Gamma \vdash c : S_2} \quad \text{CONJ_EL2} \\
\frac{\Gamma \vdash c_1 : S_1 \quad \Gamma \vdash c_2 : S_2}{\Gamma \vdash c_1 \times c_2 : S_1 || S_2} \quad \text{PAR_IN} \\
\frac{\Gamma \vdash c_1 : S_1 \quad \Gamma \vdash c_2 : (S_1, S_2)}{\Gamma \vdash c_1 \times c_2 : S_2} \quad \text{CONT_EL} \\
\frac{\Gamma \vdash c : S_1 \sqcap S_2 \quad \Gamma \vdash S_2 \sqsubseteq S_3}{\Gamma \vdash c : S_2 \sqcap S_3} \quad \text{CRE} \\
\frac{\Gamma \vdash c : S_1 \sqcap S_3 \quad \Gamma \vdash S_1 \sqsubseteq S_2}{\Gamma \vdash c : S_2 \sqcap S_3} \quad \text{CR} \\
\frac{\Gamma \vdash P_1 \quad \Gamma \vdash P_2}{\Gamma \vdash P_1 \wedge P_2} \quad \text{AND_IN} \\
\frac{\Gamma \vdash P_1 \wedge P_2}{\Gamma \vdash P_1} \quad \text{AND_EL1} \\
\frac{\Gamma \vdash P_1 \wedge P_2}{\Gamma \vdash P_2} \quad \text{AND_EL2} \\
\frac{\Gamma, P_1 \vdash P_2}{\Gamma \vdash P_1 \rightarrow P_2} \quad \text{IMP_IN} \\
\frac{\Gamma \vdash P_1 \rightarrow P_2 \quad \Gamma \vdash P_1}{\Gamma \vdash P_2} \quad \text{IMP_EL} \\
\frac{\Gamma \vdash P_1}{\Gamma \vdash P_1 \vee P_2} \quad \text{OR_IN1} \\
\frac{\Gamma \vdash P_2}{\Gamma \vdash P_1 \vee P_2} \quad \text{OR_IN2}
\end{array}$$

$$\begin{array}{c}
\Gamma \vdash P_1 \vee P_2 \\
\Gamma, P_1 \vdash P \\
\Gamma, P_2 \vdash P \\
\hline
\Gamma \vdash P \quad \text{OR_EL}
\end{array}$$

$$\begin{array}{c}
\Gamma, P_1 \vdash P_2 \\
\Gamma, P_1 \vdash \neg P_2 \\
\hline
\Gamma \vdash \neg P_1 \quad \text{NOT_IN}
\end{array}$$

$$\begin{array}{c}
\Gamma \vdash \neg \neg P \\
\hline
\Gamma \vdash P \quad \text{NOT_EL}
\end{array}$$